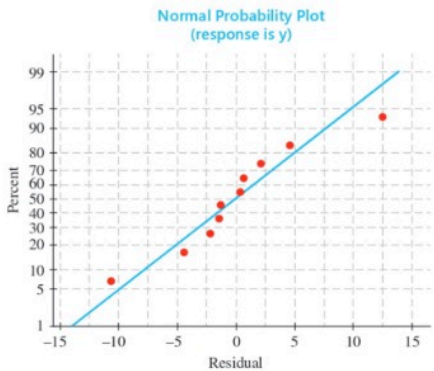
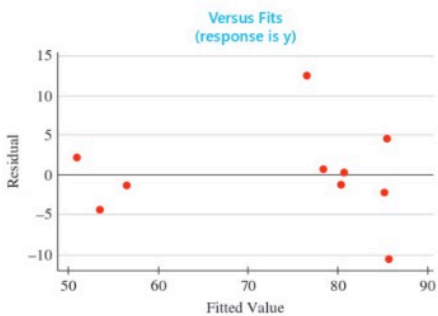
劉耀徳 112700026

5. Old Faithful The waiting time between eruptions of Old Faithful geyser in Yellowstone National Park (y) depends upon the length of time of the last eruption (x). The residual plots that follow are based on the following data.[10]

x	4.317	3.967	4.7	4.283	2.267	4.716	2.017	4.117	4.733	1.8
y	81	89	83	79	55	90	49	79	75	53





Is there anything unusual about either of these plots? Does it appear that any regression assumptions have been violated?

The plots show no serious violations of regression assumptions.
The residuals are approximately normal and randomly scattered.
However, there are a few possible outliers.

8. Laptops and Learning, again Refer to the data given in Exercise 16 (Section 12.3). The *MINITAB* printout is reproduced here.

Regression Analysis: y versus x

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	3254.03	3254.03	56.05	0.000
Error	18	1044.92	58.05		
Total	19	4298.95			

Model Summary

S	R-sq	R-sq(adj)
7.61912	75.69%	74.34%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value
Constant	−26.8	14.8	−1.82	0.086
x	1.262	0.169	7.49	0.000

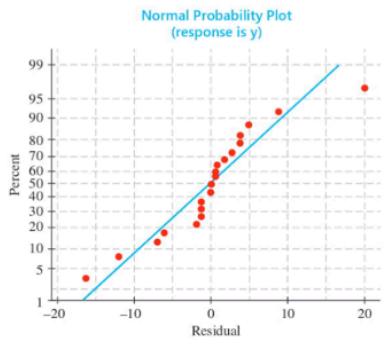
Regression Equation

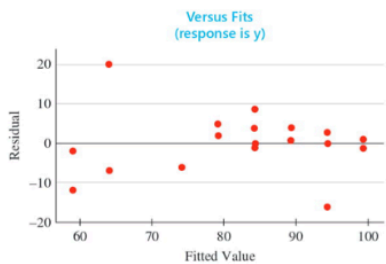
$y = -26.8 + 1.262\,x$

a. What assumptions must be made about the distribution of the random error ε?

b. What is the best estimate of σ^2, the variance of the random error ε?

c. Use the diagnostic plots for these data to comment on the validity of the regression assumptions.





(a) $\varepsilon \overset{iid}{\sim} N(0, \sigma^2)$

(b) The estimate of σ^2 is 58.05

(c) No serious violations, but these outliers may affect the model

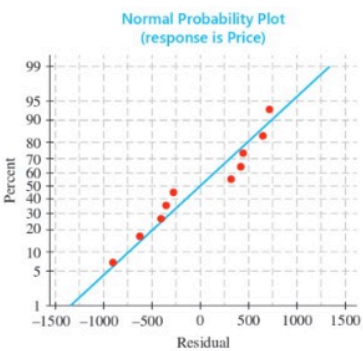
DATA SET DS1218

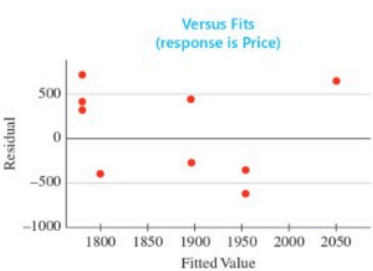
9. How to Choose a TV *Consumer Reports*[11] gave the prices and screen sizes for the top 10 LCD TVs in the 46-inch and higher categories. Does the price of an LCD TV depend on the size of the screen?

Brand	Price ($)	Size
Sony Bravia KDL-52NX800	2340	52
Samsung LN55C650	1600	55
Vizio VF550M	1330	55
Sony Bravia KDL-60EX700	2700	60
Sharp Aquos LED LC-52LE700UN	1620	52
Sony Bravia KDL-46XBR10	2500	46
Samsung UN46C8000	2200	46
Vizio SV472XVT	1400	47
Samsung UN46C7000	2100	46
LG 47LD450	900	47

a. Assuming that the relationship between size and price is linear, we perform a linear regression, resulting in a value of $r^2 = .027$. What does the value of r^2 tell you about the strength of the relationship between price and screen size?

b. The diagnostic plots for this data are shown below. Does it appear that either the normality or equal variance assumptions have been violated?





c. Use a scatterplot to plot price versus screen size for the 10 LCD TVs. Based on the information in part a, which assumption for the linear regression model has been violated?

(a) $r^2 = 0.027$ means only 2.7% of the variation in price is explained by screen size, so the relationship is very weak.

(b) The normality and equal variance assumption do not appear to be seriously violated.

(c) The linearity assumption appears to be violated

DATA SET DS1219

12. What to Buy? A marketing research experiment was conducted to study the relationship between the length of time necessary for a buyer to reach a decision and the number of alternative package designs of a product presented. The products were identical except for the package design. The length of time necessary to reach a decision was recorded for 15 participants in the marketing research study.

Length of Decision Time, y (sec)	5, 8, 8, 7, 9	7, 9, 8, 9, 10	10, 11, 10, 12, 9
Number of Alternatives, x	2	3	4

a. Find the least-squares line appropriate for these data.

b. Plot the points and graph the line as a check on your calculations.

c. Calculate s^2.

d. Do the data present sufficient evidence to indicate that the length of decision time is linearly related to the number of alternative package designs? (Test at the $\alpha = .05$ level of significance.)

e. Find the approximate p-value for the test and interpret its value.

f. If they are available, examine the diagnostic plots to check the validity of the regression assumptions.

g. Estimate the average length of time necessary to reach a decision when three alternatives are presented, using a 95% confidence interval.

$x_i y_i = 411 \qquad x_i^2 = 145$

$\rightarrow n = 15 \qquad y_i^2 = 1204$

(a) $\bar{x} = \frac{45}{15} = 3$

$\bar{y} = \frac{132}{15} = 8.8$

$\hat{y} = a + bx$

$b = \frac{S_{xy}}{S_{xx}}$

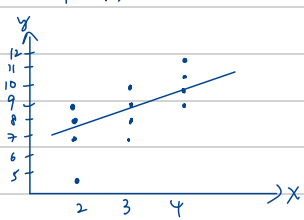
$= \frac{15}{10} = 1.5$

$a = \bar{y} - b\bar{x}$

$= 8.8 - 1.5(3) = 4.3$

$\therefore \hat{y} = 4.3 + 1.5x$

(b)



(c) $SSE = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 42.4 - \frac{15^2}{10} = 19.9$

$MSE = \frac{SSE}{n-2} = \frac{19.9}{13} = 1.5038$

(d) $H_0: \beta_1 = 0 \qquad H_a: \beta_1 \neq 0$

$t = \frac{\hat{\beta}_1 - 0}{\sqrt{\frac{MSE}{S_{xx}}}} = \frac{1.5}{\sqrt{\frac{1.5038}{10}}} = 3.87$

$\because RR = \{T > t_{13, 0.025} = 2.1604\}$

$\therefore t \in RR$, Reject H_0

(e) p-value $= P(T > 3.87) \approx 0 < 0.05$

$\therefore$ reject H_0

(f) No serious violations of regression assumptions

(g) $x = 3$

$\hat{y} = 4.3 + 1.5(3)$

$= 8.8$

95% C.I. for $\hat{y}$

$= \hat{y} \pm t_{13, 0.025} \sqrt{MSE\left(\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}\right)}$

$= 8.8 \pm 2.16 \sqrt{1.5038\left(\frac{1}{15} + 0\right)}$

$= 8.8 \pm 0.684$

$= (8.116, 9.484)$

DATA SET DS1221

15. Strawberries III The following data (Exercise 16, Section 12.2) were obtained in an experiment relating the dependent variable y (texture of strawberries) with x (coded storage temperature).

x	−2	−2	0	2	2
y	4.0	3.5	2.0	0.5	0.0

a. Estimate the expected strawberry texture for a coded storage temperature of $x = -1$. Use a 99% confidence interval.

b. Predict the particular value of y when $x = 1$ with a 99% prediction interval.

c. At what value of x will the width of the prediction interval for a particular value of y be a minimum, assuming n remains fixed?

$\bar{x} = 0$ $\quad X_iY_i = -14$

$\bar{y} = 2$

$\hat{y} = a + bx$

$b = \frac{S_{xy}}{S_{xx}} = \frac{-14}{16} = -0.875$ $\quad a = \bar{Y} - b\bar{X} = 2$

$\therefore \hat{y} = 2 - 0.875x$ $\quad SSE = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 0.25$

(a) $X = -1$

$\hat{y} = 2 - 0.875(-1)$

$= 2.875$

$MSE = \frac{SSE}{n-2} = \frac{0.25}{3} = 0.083$

99% C.I. for $\hat{y} = \hat{y} \pm t_{3,0.005}\sqrt{MSE\left(\frac{1}{n} + \frac{(X-\bar{X})^2}{S_{xx}}\right)}$

$= 2.875 \pm 5.84\sqrt{0.083\left(\frac{1}{5} + \frac{1}{16}\right)}$

$= 2.875 \pm 0.86$

$= (2.015, 3.735)$

(b) $X = 1$

$\hat{y} = 2 - 0.875(1)$

$= 1.125$

99% P.I. for $\hat{y} = 1.125 \pm 5.84\sqrt{0.083\left(1 + \frac{1}{5} + \frac{1}{16}\right)}$

$= 1.125 \pm 1.89$

$= (-0.765, 3.015)$

(c) Minimum width occurs at $x = 0$

DATA SET DS1223

18. Plant Science An experiment was conducted to determine the effect of various levels of phosphorus on the inorganic phosphorus levels in Sudan grass. The data in the table represent the levels of inorganic phosphorus in micromoles (μmol) per gram dry weight. Use the *MINITAB* output to answer the questions.

Phosphorus Applied, x	Phosphorus in Plant, y
.50 μmol	204
	195
	247
	245
.25 μmol	159
	127
	95
	144
.10 μmol	128
	192
	84
	71

a. Plot the data. Do the data appear to exhibit a linear relationship?

b. Find the least-squares line relating the plant phosphorus levels y to the amount of phosphorus applied to the soil x. Graph the least-squares line as a check on your answer.

c. Do the data provide sufficient evidence to indicate that the amount of phosphorus present in the plant is linearly related to the amount of phosphorus applied to the soil?

d. Estimate the mean amount of phosphorus in the plant if .20 μmol of phosphorus is applied to the soil. Use a 90% confidence interval.

Regression Analysis: y versus x

Coefficients

Term	Coef	SE Coef	T-Value	P-Value
Constant	80.85	22.40	3.61	0.005
x	270.82	68.31	3.96	0.003

Regression Equation

y = 80.9 + 270.8 x

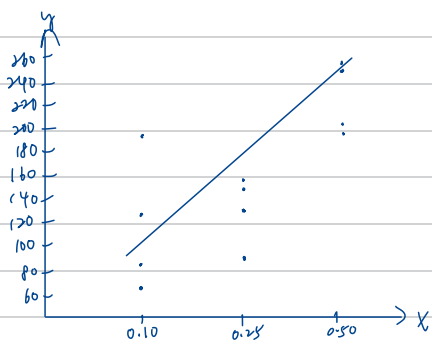
Settings

Variable	Setting
x	0.2

Prediction

Fit	SE Fit	90% CI	90% PI
135.015	12.6264	(112.130, 157.900)	(60.6448, 209.386)

(a)



⇒ positive linear relationship

(b) $b_0 = 80.85$, $b_1 = 270.82$

$\therefore \hat{y} = 80.85 + 270.82x$

(c) $H_0: \beta_1 = 0$, $H_a: \beta_1 \neq 0$

$p = 0.003$

$\because 0.003 < 0.05$

$\therefore$ Reject H_0

(d) $x = 0.20$

$\hat{y} = 80.85 + 270.82(0.20)$

$= 135.014$

90% C.I. = (112.130, 157.900)

DATA SET DS1227

13. Geothermal Power Geothermal power is an important source of energy. Since the amount of energy contained in 1 pound of water is a function of its temperature, you might wonder whether water obtained from deeper wells contains more energy per pound. The data in the table are reproduced from an article on geothermal systems.[17]

Location of Well	Average (max.) Drill Hole Depth (m) x	Average (max.) Temperature (°C) y
El Tateo, Chile	650	230
Ahuachapan, El Salvador	1000	230
Namafjall, Iceland	1000	250
Larderello (region), Italy	600	200
Matsukawa, Japan	1000	220
Cerro Prieto, Mexico	800	300
Wairakei, New Zealand	800	230
Kizildere, Turkey	700	190
The Geysers, United States	1500	250
	8050	2100

Is there a significant positive correlation between average maximum drill hole depth and average maximum temperature?

$S_{xy} = \sum X_i Y_i - n\bar{X}\bar{Y}$

$= 23166.67$

$n = 9$

$S_{xx} = \sum X_i^2 - n\bar{X}^2$

$= 602222.22$

$S_{yy} = \sum Y_i^2 - n\bar{Y}^2$

$= 8200$

$H_0: \rho = 0$

$H_a: \rho \neq 0$

$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$ $\quad r^2 = 0.1082$

$= 0.329$

$t = r \cdot \sqrt{\frac{n-2}{1-r^2}}$

$= 0.329 \sqrt{\frac{7}{0.8918}}$

$= 0.92$

$\because RR = \{ t > t_{7, 0.05} = 1.89469 \}$

$\therefore t \notin RR$, Non-reject H_0

DATA SET DS1228

14. Ice Cream, Anyone? The popular ice cream franchise *Coldstone Creamery* posted the nutritional information for its ice cream offerings in three serving sizes—"Like it," "Love it," and "Gotta Have it"—on their website.[18] A portion of that information for the "Like it" serving size is shown in the table.

Flavor	Calories	Total Fat (grams)
Cake Batter	340	19
Cinnamon Bun	370	21
French Toast	330	19
Mocha	320	20
OREO® Crème	440	31
Peanut Butter	370	24
Strawberry Cheesecake	320	21
	2490	155

a. Should you use the methods of linear regression analysis or correlation analysis to analyze the data? Explain.

b. Analyze the data to determine the nature of the relationship between total fat and calories in *Coldstone Creamery* ice cream.

$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y}$

$= 56140 - 7(355.71)(22.14)$

$= 1012.06$

$S_{xx} = \sum x_i^2 - n\bar{x}^2$

$= 10992.77$

$S_{yy} = \sum y_i^2 - n\bar{y}^2$

$= 109.74$

(a) Use correlation analysis

(b) $r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$

$= 0.92$

$\therefore$ strong positive relationship

DATA SET DS1229

15. Body Temperature and Heart Rate Is there any relationship between these two variables? To find out, we randomly selected 12 people from a data set constructed by Allen Shoemaker (*Journal of Statistics Education*) and recorded their body temperature and heart rate.[19]

Person	1	2	3	4	5	6
Temperature (degrees)	96.3	97.4	98.9	99.0	99.0	96.8
Heart Rate (beats per minute)	70	68	80	75	79	75

Person	7	8	9	10	11	12
Temperature (degrees)	98.4	98.4	98.8	98.8	99.2	99.3
Heart Rate (beats per minute)	74	84	73	84	66	68

a. Find the correlation coefficient r, relating body temperature to heart rate.

b. Is there sufficient evidence to indicate that there is a correlation between these two variables? Test at the 5% level of significance.

$X = 1180.3$

$Y = 896$

$n = 12$

$S_{xy} = \sum X_i Y_i - n\bar{X}\bar{Y}$

$= 88940.6 - 12(98.3583)(74.6667)$

$= 11.53$

$S_{xx} = \sum X_i^2 - n\bar{X}^2$

$= 116103.03 - 12(98.3583)^2$

$= 10.69$

$S_{yy} = \sum Y_i^2 - n\bar{Y}^2$

$= 67312 - 12(74.6667)^2$

$= 410.67$

(a) $r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}$

$= 0.174$

(b) $H_0 : \rho = 0$

$H_a : \rho \neq 0$

$t = r \cdot \sqrt{\frac{n-2}{1-r^2}}$

$= 0.174 \sqrt{\frac{10}{1-0.174^2}}$

$= 0.559$

$\because RR = \{ t_{10, 0.025} = 2.2281 \}$

$\therefore t \notin RR$, Non-reject H_0